



PROJECT 3: DATA SPRINT

Group : HDBee

Jimmy Myo Myint Aung

Ying Weng Seng

Nathaniel Tan

Shadni Wardiah

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AGENDA

PROJECT OVERVIEW

Background & Problem Statement
Objectives
Data Sources

DATA PRESENTATION & ANALYSIS

Exploratory Analysis (EDA)
Feature Engineering
Insights from the Data

MODELING & RESULTS

Correlation Analysis – What Drives Price?
Model Overview
Model Comparison
What Singaporeans Really Value in an HDB Home?

IMPACT & LESSONS

Assumptions & Caveats
Business Impact & Recommendations
Lessons Learned & Next Steps
Retrospective

01

02

03

04

PROJECT OVERVIEW





BACKGROUND

- Similar HDB flats (same town/size) can differ by S\$100k+ (e.g., S\$400k vs S\$550k).
- Key question: is the gap from floor level, MRT/schools, flat age, or market timing?
- Upfront, explainable pricing helps with faster decisions and fair negotiations.
- We will focus on observable drivers (e.g. flat attributes & location), not opaque or hard-to-interpret features.

Team goal: turn raw transactions into simple, defensible insights for everyday use.



PROBLEM STATEMENT

- HDB resale prices are dynamic and competitive, shifting with buyer preferences, economic conditions, and location demand.
- WOW! agents need to deliver accurate, competitive price estimates across all flat types and neighborhoods.
- Without strong guidance, sellers risk undervaluing and buyers risk overpaying.
- Our goal is to build a transparent, data-driven model to support fair pricing and confident client advice.

OBJECTIVES

IDENTIFY KEY PRICE DRIVERS

- Analyze how variables such as flat size, age, location, and proximity to amenities (MRT, schools, malls, hawker centres) influence resale prices

COMPARE PRICING PATTERNS

- Examine price variations between neighborhoods and flat models to highlight high-value and emerging areas.

DEVELOP A PREDICTIVE MODEL FOR RESALE PRICE ESTIMATION

- Build and test data-driven models to estimate resale prices based on key property attributes

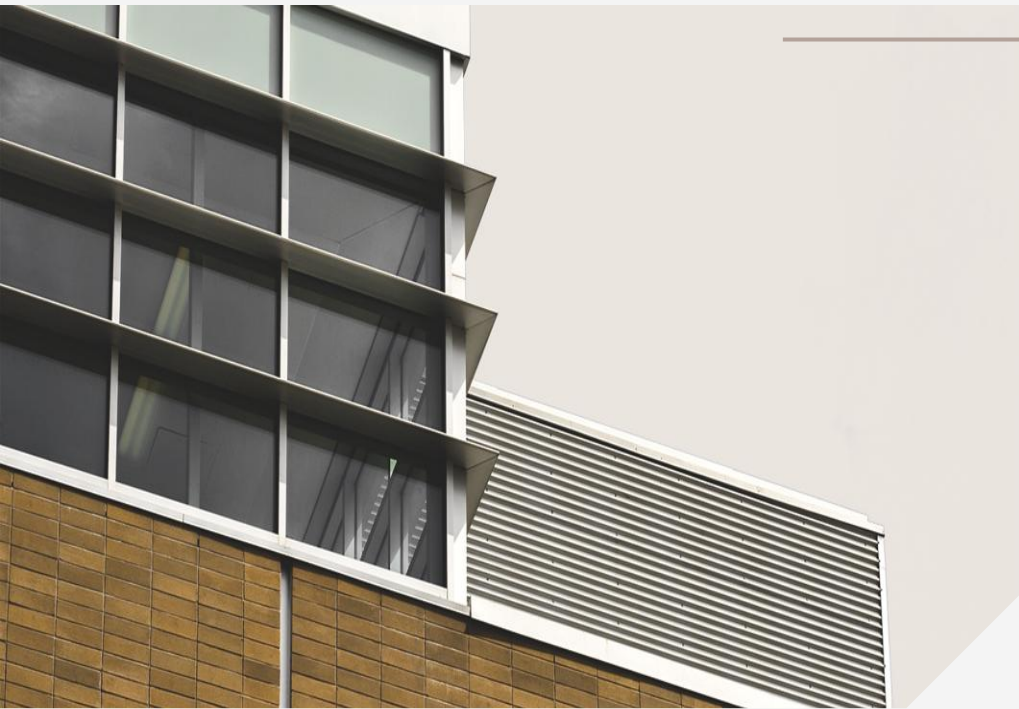
VISUALIZE INSIGHTS FOR DECISION-MAKING

- Create interactive dashboards and visual tools to help agents and clients easily interpret price trends and factors

SUPPORT BUSINESS STRATEGY AND CLIENT ADVISORY

- Use analytical findings to enhance client consultations, improve pricing accuracy, and strengthen WOW!'s market positioning

DATA SOURCES



DATASET

train.csv -- this data contains all of the training data for our model.

test.csv -- this data contains the test data for our model.

sample_sub_reg.csv -- An example of a correctly formatted submission to evaluate the model accuracy on the test data.

DATA PRESENTATION & ANALYSIS



EXPLORATORY DATA ANALYSIS

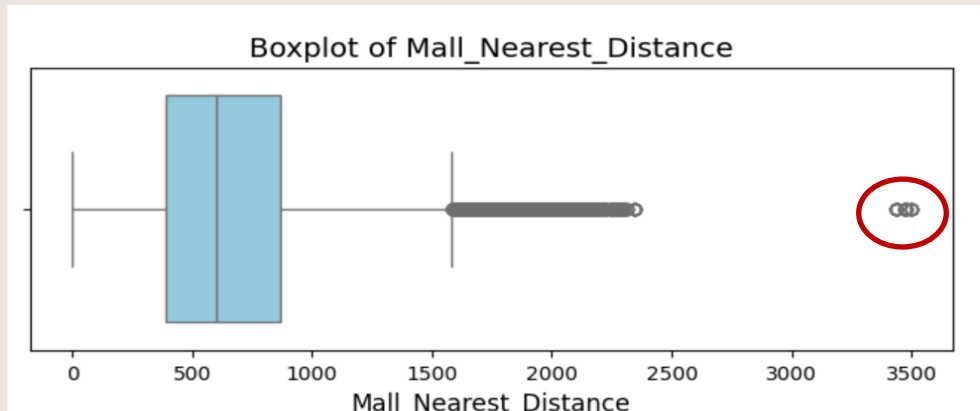
Dataset of 147,976 records and 49 variables

DATA CLEANING

- Identifying duplicates
- Identifying nulls (7 columns)
 - Mall_Nearest_Distance
 - Mall_Within_500m
 - Mall_Within_1km
 - Mall_Within_2km
 - Hawker_Within_500m
 - Hawker_Within_1km
 - Hawker_Within_2km
- Verifying data types
- Checking outliers

DATA HANDLING

- No duplicates found
- Replace nulls with 0 for 'Mall_Within_XX km' and 'Hawker_Within_XX km'
- Removing rows where by Mall_Nearest_Distance are nulls (829 rows)
- Changing data types (date, Boolean) for columns needed
- Remove outliers (Skewed high values)



Example of identifying skewed outliers

FEATURE ENGINEERING

(+) ADDED NEW FEATURES

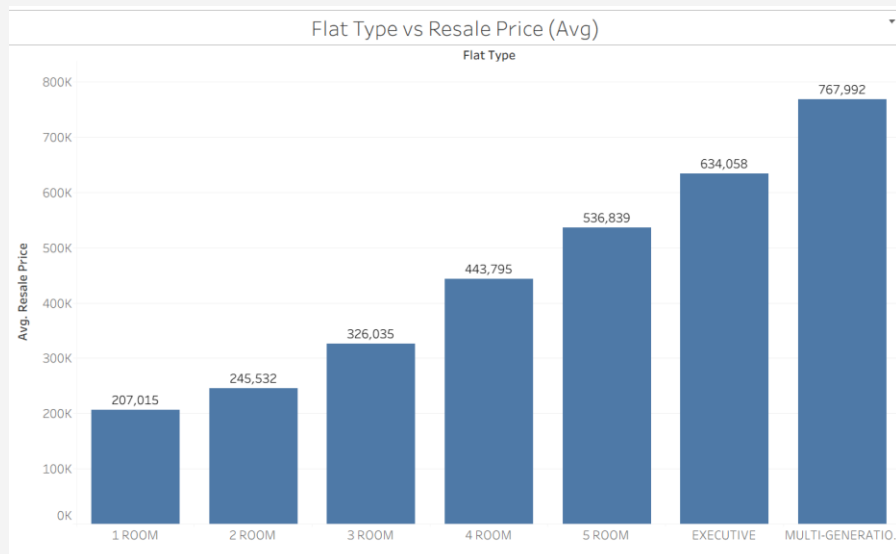
- floor_level_range
 - Floor categorised based of flat max floor
- years_of_lease_left
 - 99 years – HDB age
- mrt_region
 - Classify MRT names into country regions based to URA region guidelines

	mid_storey	max_floor_lvl	floor_level_range
0	11	25	Mid Lower Level
1	8	9	Upper Level
2	14	16	Upper Level
3	3	11	Mid Lower Level
4	2	4	Mid Upper Level
...	...	...	...
150629	5	12	Mid Lower Level
150630	5	16	Mid Lower Level
150631	11	15	Mid Upper Level
150632	8	10	Upper Level
150633	8	12	Mid Upper Level

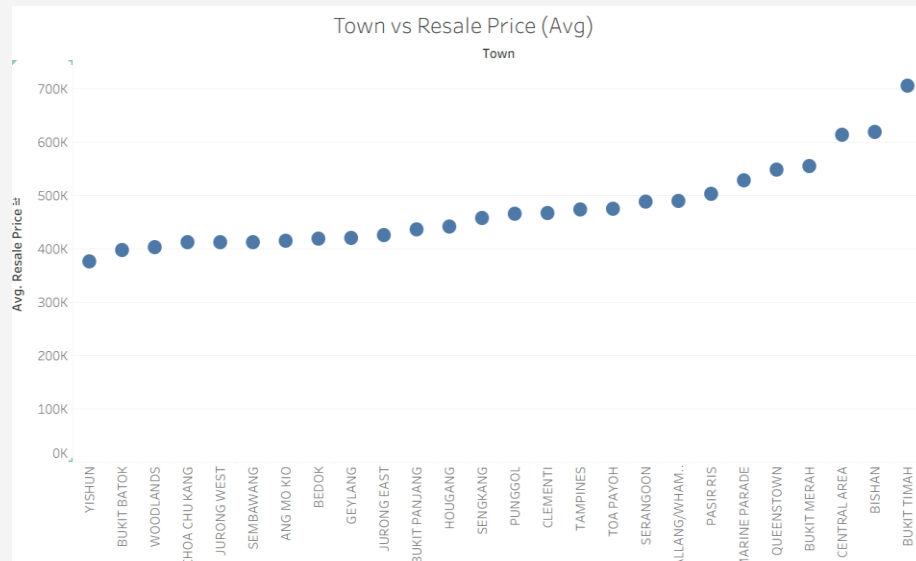
INSIGHTS FROM THE DATA



Size matters



Location/Town

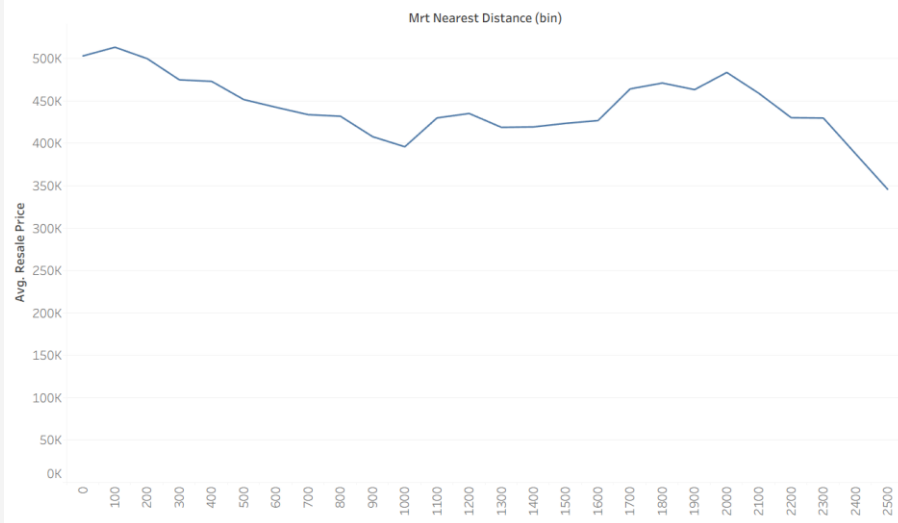


INSIGHTS FROM THE DATA



Amenities distance

Mrt Nearest Distance vs Price



Age of HDB

Lease Commence Date vs Resale Price (Avg)



MODELING & RESULTS



(+) CROSS CHECK ON CORRELATIONS BTW FEATURES

(-) CROSS CHECK ON CORRELATIONS BTW TARGET & FEATURES

CORRELATION ANALYSIS

	resale_price
floor_area_sqft	0.65669984
commercial	-0.1408056
market_hawker	-0.0059146
multistorey_carpark	-0.0077384
Mall_Within_500m	0.08871682
Mall_Within_1km	0.08580538
Mall_Within_2km	0.11424498
Hawker_Within_500m	0.01278653
Hawker_Within_1km	0.09213981
Hawker_Within_2km	0.18514101
mrt_nearest_distance	-0.1274085
mrt_interchange	0.06455517
pri_sch_nearest_distance	-0.0072261
years_of_lease_left	0.35018899
2 ROOM	-0.1605935
3 ROOM	-0.5061808
4 ROOM	-0.0245584
5 ROOM	0.34785892
EXECUTIVE	0.38477496
Mid Lower Level	0.0087973
Mid Upper Level	0.015815
Upper Level	0.03554236
east	0.02128902
north	-0.1741129
north_east	0.009977
west	-0.1028206

(1) CORRELATION BETWEEN FEATURES

- Mall/ Hawker within [500m | 1km | 2km]

(2) CORRELATION BETWEEN FEATURES & TARGET

- Market Hawker
- Multistorey Carpark
- Pri School Nearest Distance

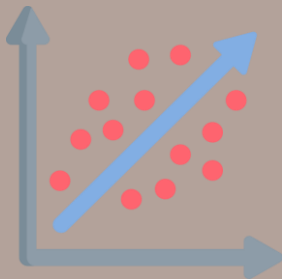
} Almost close to 0 with no correlation

MODEL OVERVIEW

MODELS USED

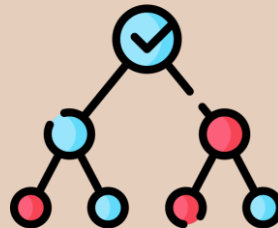
LINEAR REGRESSION MODEL

"If one factor increases, how much does the price change?"



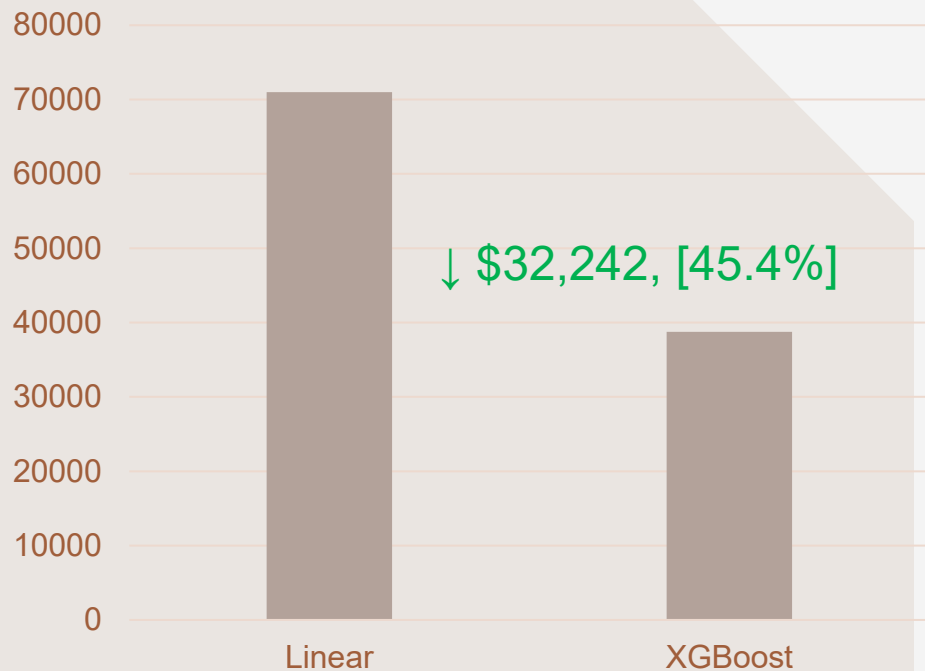
XGBOOST

Instead of one line, it builds *hundreds of small trees* that learn from each other's mistakes.

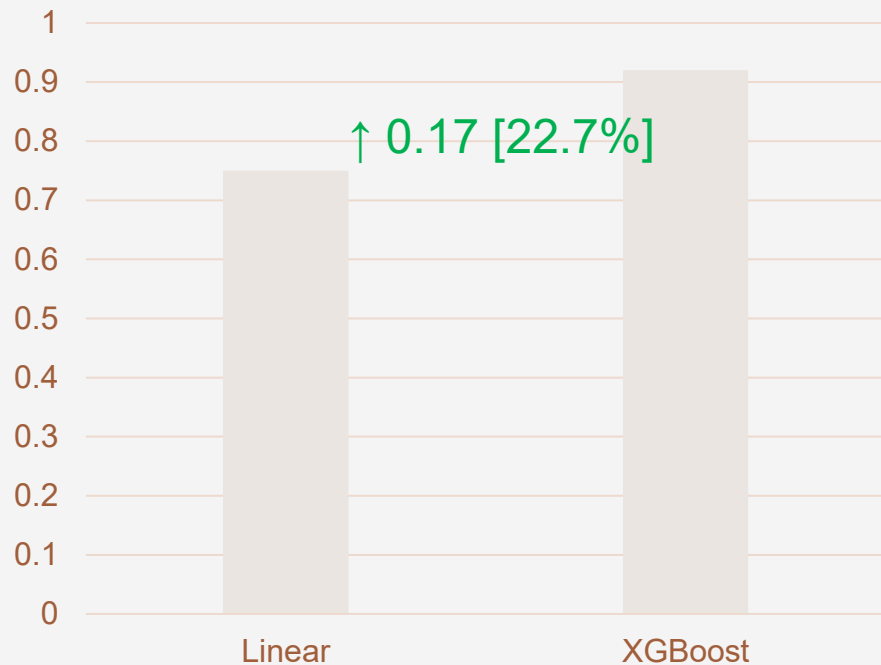


MODEL COMPARISON

RMSE

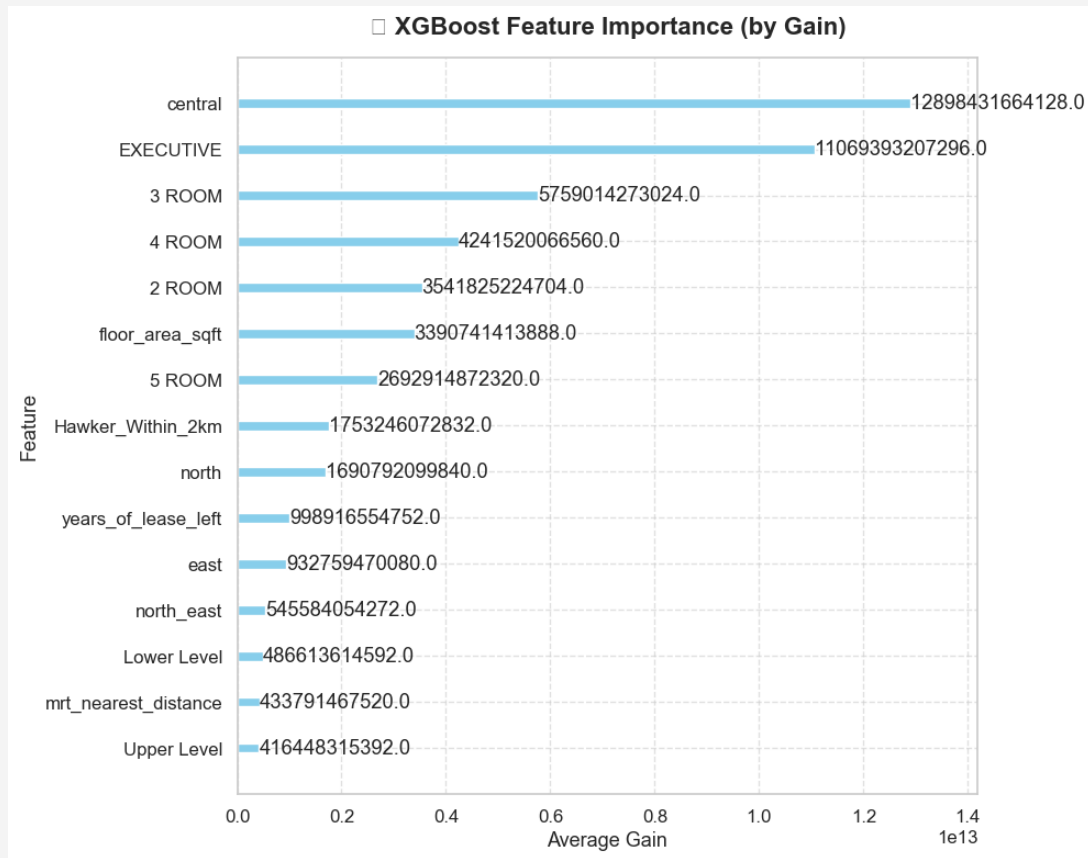


R^2



What Singaporeans Really Value in an HDB Home?

1. Central is everything
2. **BIGGER** the better
3. Proud Foodies
4. Practical
5. Everyday Conveniences



IMPACT & LESSONS



ASSUMPTIONS & CAVEATS



Assumptions



Market behaves rationally – resale prices reflect measurable factors.



Data sources are reliable and updated.



Past trends represent near-term market patterns.



Caveats



Uncaptured factors – interior condition and renovation not included.



Approximate proximity – map data, not travel time.



Market shifts – sudden policy or sentiment changes may affect predictions.

BUSINESS IMPACT & RECOMMENDATIONS



Data Insights

Enables fair and transparent pricing for both buyers and sellers.



Model Application

Predictive model allows accurate resale price estimation, reducing guesswork and bias.



Agent Empowerment

Provides data-backed confidence in negotiations, boosting agent credibility



Business Impact

Strengthens WOW!'s position as a trusted, analytics-driven real estate brand.



RECOMMENDATIONS



Integrate model into agent workflow through a Streamlit-based pricing app.



Continuously update data with new transactions to maintain accuracy.

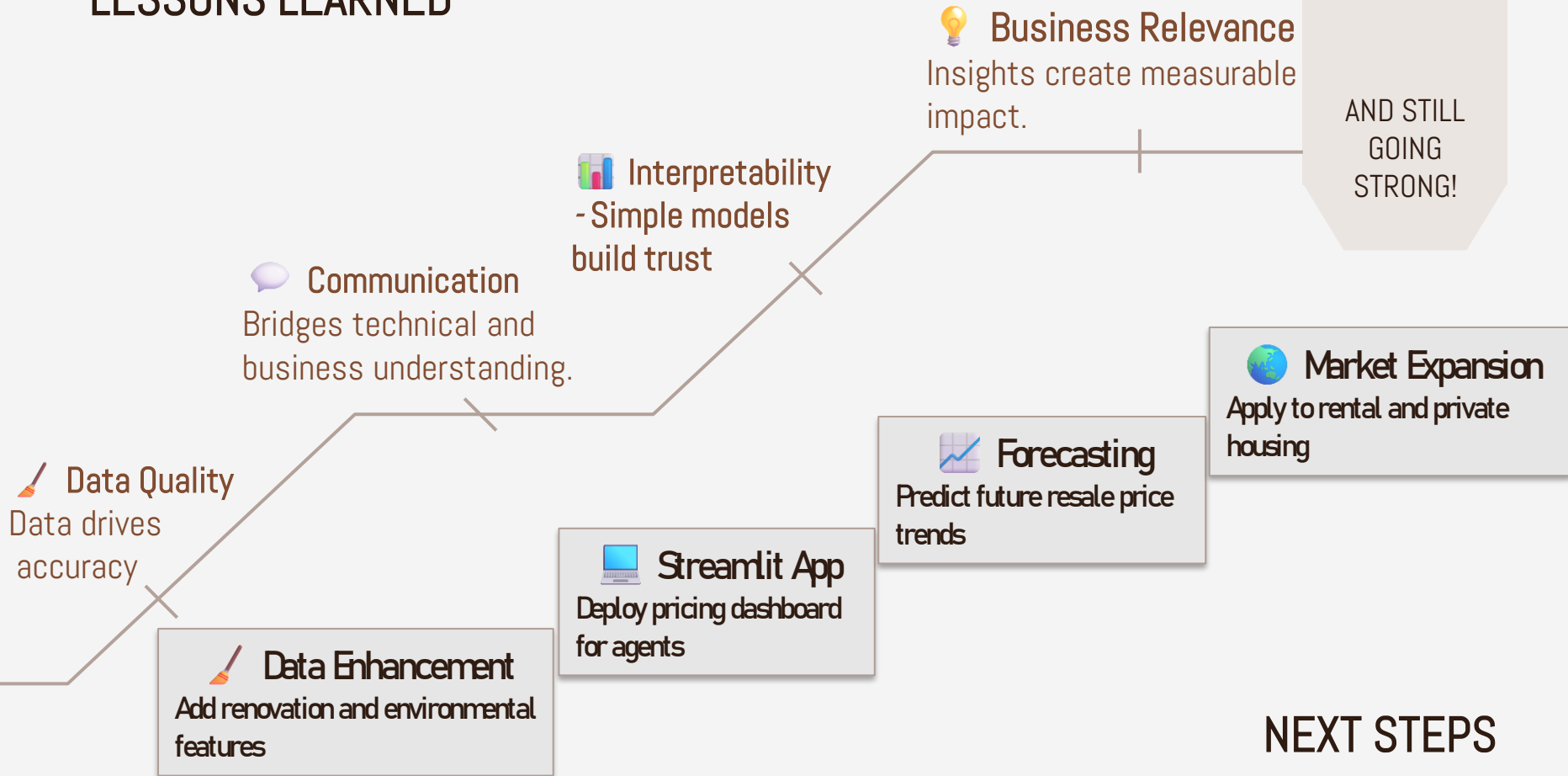


Use results for targeted marketing (e.g., towns with rising demand).



Explore rental and BTO market extensions.

LESSONS LEARNED



RETROSPECTIVE

From messy data to actionable insights — our journey combined teamwork, technical depth, way forward and storytelling.



TEAMWORK → What works?

Task division and collaboration



TECHNICAL DEPTH → What must be done?

Correlation and feature selection



WAY FORWARD → To move on or put a pin on it?

Exploration to implementation



STORYTELLING → How can we show/explain?

Make complex analytics insights
accessible to all audience



STREAMLIT DEMO





QUESTIONS?

A photograph of a modern building facade with a large, bold, dark brown 'THANK YOU' sign mounted on a light-colored, textured concrete wall. Below the sign, there are dark, rectangular window openings with vertical concrete pillars. The building is set against a light beige background with a large, light beige geometric shape on the right side.

THANK YOU

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